

CS & IT ENGINEERING

Digital Logic

Boolean Theorems and Logical Gates

Lecture No. 06



By- Aditya sir

Recap of Previous Lecture



Topic

XOR
XNOR



Topics to be Covered



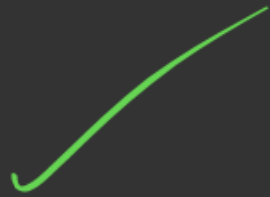
Topic

Imp Properties

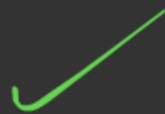
Practice Questions



XOR



XNOR



$$\overline{A \oplus B} = A \odot B$$

$$A \oplus B \oplus C = A \odot B \odot C$$

Complement Property of XOR & XNOR

$$\left(\overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B} \right) = A \odot B$$

$$\left(\overline{A \odot B} = \bar{A} \odot B = A \odot \bar{B} \right) = A \oplus B$$

$$\overline{A \oplus B \oplus C} = \overline{A} \oplus B \oplus C = A \oplus \overline{B} \oplus C = A \oplus B \oplus \overline{C}$$

$$1 \oplus 0 \oplus 1 = 0$$

$$\left\{ \overline{1 \oplus 0 \oplus 1} = \overline{0} = 1 \right.$$

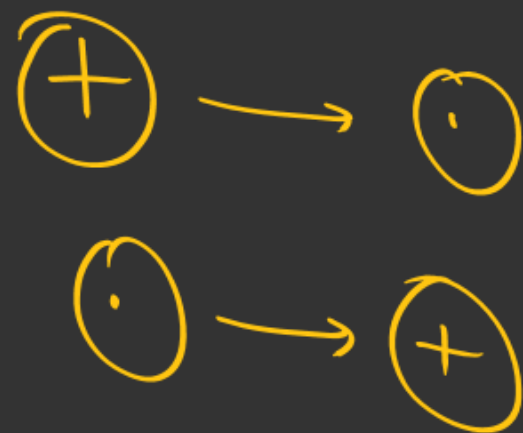
$$\left\{ \overline{1 \oplus 0 \oplus 1} = 0 \oplus 0 \oplus 1 = 1 \right.$$

$$1 \oplus \overline{0} \oplus 1 = 1 \oplus 1 \oplus 1 = 1$$

$$1 \oplus 0 \oplus \overline{1} = 1 \oplus 0 \oplus 0 = 1$$

$$\overline{A \oplus B} = \overline{A} \oplus B = A \oplus \overline{B}$$

$$\rightarrow \overline{A \oplus B} = A \odot B$$



$$\overline{A \oplus B \oplus C} = \overline{A} \oplus B \oplus C$$

$$= A \odot B \oplus C = A \oplus B \odot C$$

XOR = 1
XNOR = 1

$$A \oplus B \oplus C \odot D = A \oplus B \odot C \oplus D = A \odot B \oplus C \oplus D$$

$$\overline{1 \oplus 0 \oplus 1} = 1$$



$$= 1 \odot 0 \oplus 1 = 1$$

$$= 1 \oplus 0 \odot 0 = 1$$

$$\begin{aligned}
 \overline{\overline{A \oplus B \oplus C}} &= A \oplus B \oplus C & A \oplus B \oplus C &= \underline{A \odot B \odot C} \\
 &\downarrow & & \\
 &= \overline{A \odot B \oplus C} &= \overline{A \oplus B \odot C} & \\
 &\downarrow & & \\
 &= A \odot B \odot C &= A \odot B \odot C &
 \end{aligned}$$

$$\overline{A} \oplus \overline{B} = A \odot \overline{B} = A \oplus B$$

$$\overline{A \oplus B} \odot \overline{C} \odot \overline{D}$$

↓

$$= A \odot B \odot \overline{C} \odot \overline{D}$$

$$= A \odot B \oplus C \odot \overline{D}$$

$$= A \odot B \oplus C \oplus D$$

$$= \underline{A \oplus B \oplus C \oplus D}$$

$$\overline{A \oplus \overline{B}} \odot \overline{\overline{C} \oplus D}$$

$$\overline{A \odot B} \odot \overline{C \odot D} = \overline{A \oplus B} \odot \overline{C \oplus D}$$

$$= A \oplus B \oplus C \oplus D$$

$$\begin{aligned}
 & \overline{A \odot \bar{B} \odot \bar{C} \odot \bar{D}} \\
 & \overline{A \oplus B \odot C \oplus D} \\
 = & \overline{A \odot B \odot C \odot D} \\
 = & A \odot B \oplus C \odot D
 \end{aligned}$$

$$\begin{aligned}
 & \downarrow \quad \downarrow \\
 & \overline{A \oplus B \oplus C} \\
 & \overline{A \oplus \bar{B} \oplus C} \quad \overline{AB = P} \\
 & \quad \quad \quad \overline{P \oplus C}
 \end{aligned}$$

$$\begin{aligned}
 & \overline{A \odot B \odot \bar{C} \odot \bar{D}} \\
 & \overline{A \odot B \odot C \odot D} \\
 = & A \odot B \odot C \odot D \\
 = & A \odot B \oplus C \odot D
 \end{aligned}$$



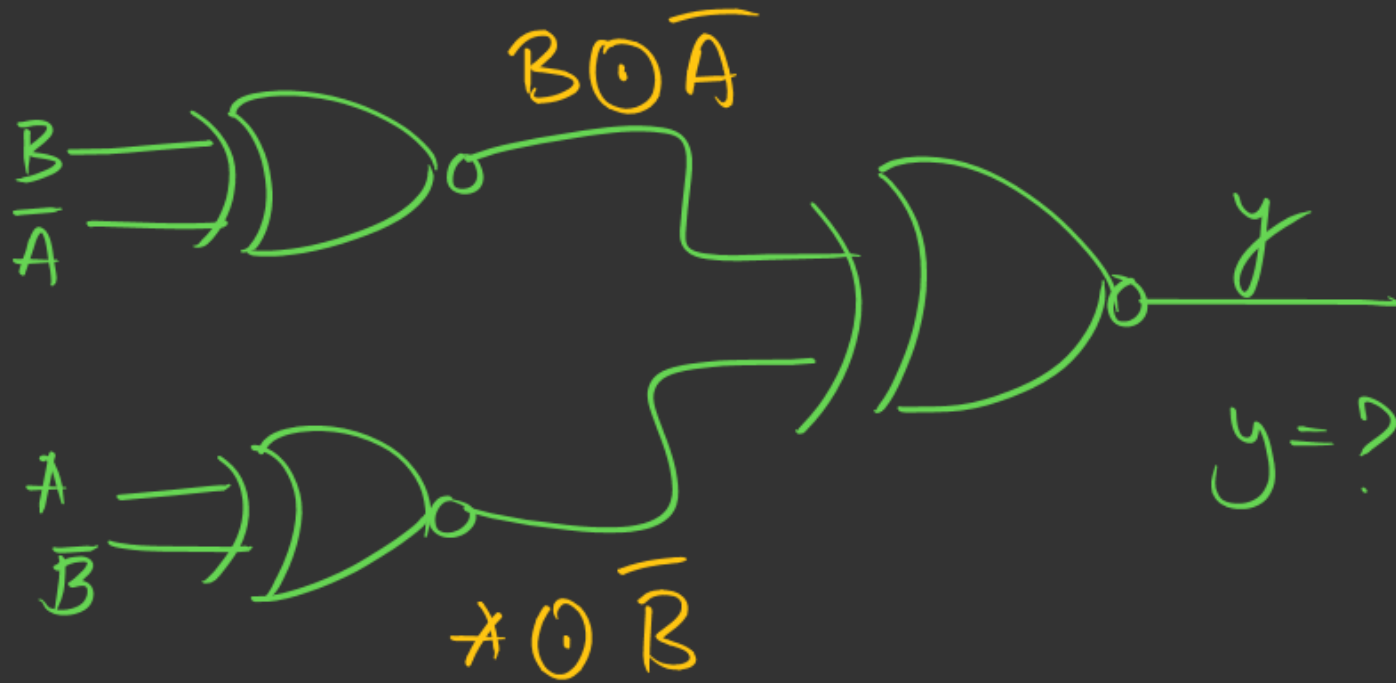
$$\overline{A \odot \overline{B \odot \overline{C \odot D \odot \overline{E}}}}$$

$$\overline{7} = 1$$

$$= (A \odot B \odot C \odot D \odot E)$$

$$= A \oplus B \oplus C \oplus D \oplus E$$

Hw) :-



$$A \oplus B = P$$

$$(B \oplus \bar{A}) \oplus (A \oplus \bar{B})$$

$$(A \oplus B) \oplus (A \oplus B) \rightarrow P \oplus P = \underline{\underline{1}}$$

HW2:- $y = \overline{\overline{A \oplus B} \oplus \overline{C}} = \overline{A \odot B \odot C}$
 $= A \odot B \oplus C$

A) $A \odot B \oplus C$

X B) $\overline{\overline{A \oplus B} \odot C} \Rightarrow \overline{A \odot B \oplus C} = A \oplus B \oplus C$

X C) $\overline{A \oplus B \oplus \overline{C}} \rightarrow \overline{A \oplus B \odot C} = A \oplus B \oplus C$

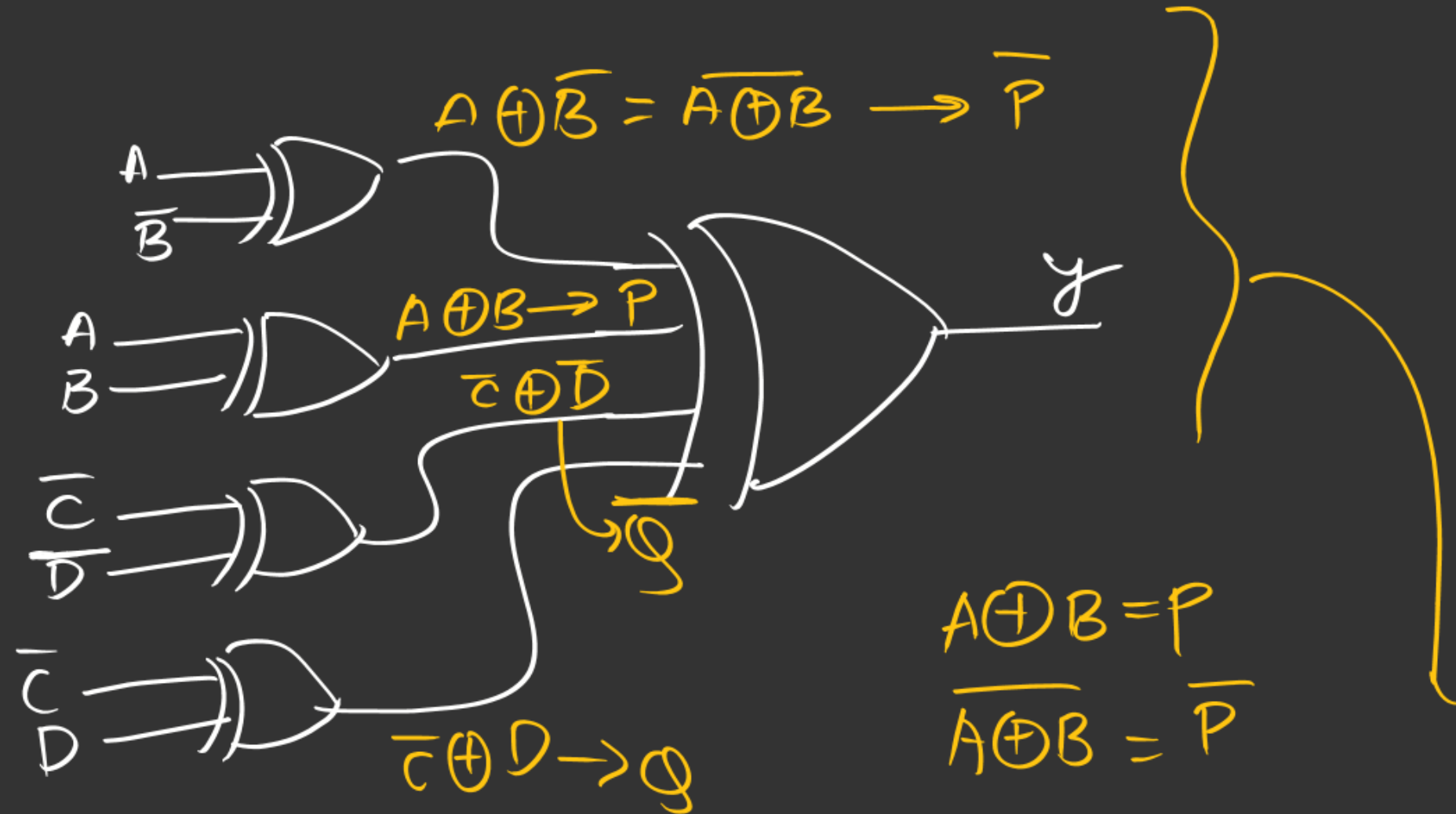
X D) $A \oplus B \oplus C$

A) $A \odot B \oplus C \oplus D$

B) $A \oplus B \oplus C \oplus D$

C) $A \odot B \odot C \odot D$

~~D) Always 0~~



$A \oplus B = P$

$\overline{A \oplus B} = \bar{P}$

$\bar{C} \oplus D = Q$

$\bar{C} \oplus \bar{D} = \bar{Q}$

$$Y = \bar{P} \oplus P \oplus Q \oplus \bar{Q}$$

$$= \underline{\underline{1}} \oplus \underline{\underline{1}} = 0$$

$$(8) \quad y = A \oplus (A+B)$$

$$A) A \oplus B$$

$$B) y = A \odot B$$

$$C) \bar{A}B$$

$$D) A\bar{B}$$

$$P \oplus Q = \bar{P}Q + P\bar{Q}$$

$$A \oplus (A+B)$$

$$= \bar{A} \cdot (A+B) + A \cdot (\overline{A+B})$$

$$= \bar{A} \cdot A + \bar{A}B + A(\bar{A} \cdot \bar{B})$$

$$= 0 + \bar{A}B + A\bar{A}\bar{B}$$

$$= \bar{A}B$$

[XOR & XNOR] $\rightarrow$ not distributive over AND
and over OR operator

$$Y = A \oplus (A + B) \neq (A \oplus A) + (A \oplus B)$$
$$0 + A \oplus B$$
$$= A \oplus B$$

$$Y = A \oplus (B \cdot C)$$
$$\neq (A \oplus B) \cdot (A \oplus C)$$

(Q) $y = A \oplus B \oplus AB$

A) AB

B) $(A+B)$

C) $\overline{AB}$

D) $\overline{A+B}$

$$= P(P + \text{anything})$$

$$= P$$

$$y = A \oplus \underline{\underline{B \oplus AB}}$$

$$= A \oplus (\overline{B} \cdot AB + B \cdot (\overline{AB}))$$

$$= A \oplus (0 + B(\overline{A} + \overline{B}))$$

$$= A \oplus (0 + B\overline{A} + B\overline{B})$$

$$= A \oplus (B\overline{A}) = \overline{A} \cdot (B\overline{A}) + A(\overline{B\overline{A}})$$

$$= \overline{A}B + A(\overline{B} + A)$$

$$= \overline{A}B + A = (A + \overline{A})(A + B) = \underline{\underline{A + B}}$$

$$A \oplus B \oplus AB = (A+B)$$



$$A \oplus B \oplus (A+B) = AB$$

$$A \cdot (B \oplus C) = A \cdot B \oplus A \cdot C$$

$$= \bar{A}B \cdot AC + AB \cdot \bar{A}C$$

$$= (\bar{A} + B) \cdot AC + AB(\bar{A} + C)$$

$$= \cancel{A}A + \bar{B}AC + \cancel{A}B\bar{A} + AB\bar{C}$$

$$= A\bar{B}C + AB\bar{C} = A(\bar{B}C + B\bar{C})$$

$$= A(B \oplus C)$$

$$\begin{array}{l} 1) A + (B \oplus C) \neq (A + B) \oplus (A + C) \\ 2) A \cdot (B \odot C) \neq (A \cdot B) \odot (A \cdot C) \end{array}$$

$$\begin{aligned}
 & A \oplus (B \oplus AB) \\
 &= A \oplus (B(1 \oplus A)) = A \oplus (B \cdot \bar{A}) \neq A \odot B \cdot A
 \end{aligned}$$

$$\begin{aligned}
 \rightarrow & \rightarrow = \overline{\bar{A} \oplus (B \cdot \bar{A})} = \overline{\bar{A}(1 \oplus B)} \\
 & = \overline{\bar{A} \cdot \bar{B}} = (A + B)
 \end{aligned}$$

$$\overline{A} = P \rightarrow \overline{P} = A$$

$$1) A \oplus B \oplus \overline{A}B = \overline{A \oplus B \oplus \overline{A}B} = \overline{P \oplus Q \oplus PQ}$$

$$2) A \oplus B \oplus \overline{A} \cdot \overline{B}$$

$$= \overline{(P+Q)} = \overline{P} \cdot \overline{Q} = (\overline{A} \cdot \overline{B})$$

$$P = \overline{A}$$

$$Q = \overline{B}$$

$$\overline{A \oplus B \oplus \overline{A} \cdot \overline{B}}$$

$$= P \oplus Q \oplus P \cdot Q$$

$$= (P+Q) = (\overline{A} + \overline{B})$$

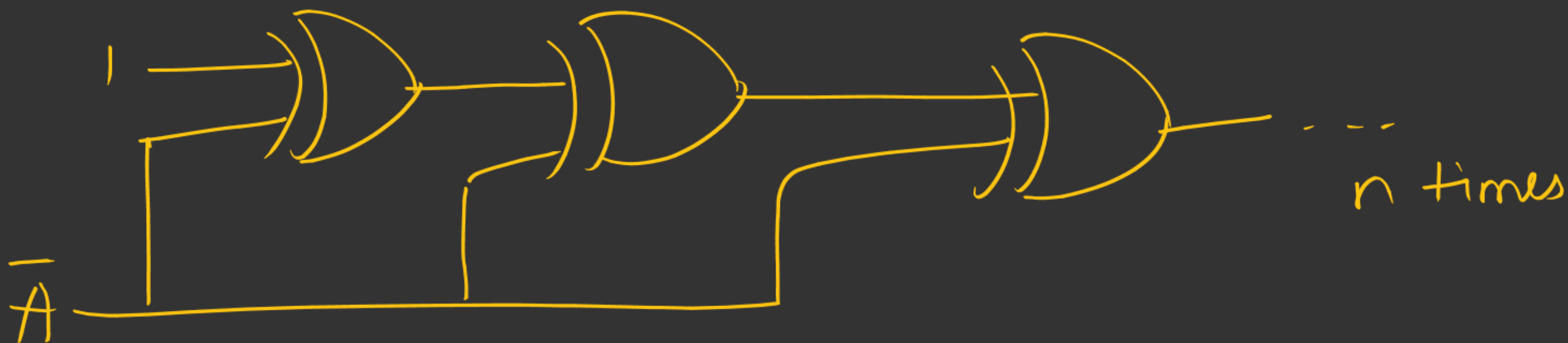
$$A \oplus B \oplus (A + \bar{B})$$

$$= \overline{A \oplus \bar{B} \oplus (A + \bar{B})} \longrightarrow = \overline{A \cdot \bar{B}} = \underline{\underline{(\bar{A} + B)}}$$

$$= P \oplus Q \oplus (P + Q) = P \cdot Q$$

HW

mcq



True?

A) A , if $n = \text{even}$

B) $\bar{A}$, if $n = \text{even}$

C) A , if $n = \text{odd}$

D) 1 , if $n = \text{odd}$



2 mins Summary



Topic

Topic

Topic

Topic

Topic

Properties
Questions



THANK - YOU